Bayesian Analysis of Bike Sharing Demand: A Poisson–Gamma Conjugate Framework

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Abstract

This report presents a Bayesian analysis of hourly bike rental demand using the UCI Bike Sharing Dataset [Fanaee-T, 2013]. We begin with a homogeneous Poisson likelihood and derive the maximum likelihood estimator in closed form. A conjugate Gamma prior yields a tractable posterior whose predictive distribution is Negative Binomial. We define an "overload" event, compute posterior predictive overload probabilities, and apply calibration techniques (Platt scaling, isotonic regression) to improve probabilistic accuracy. Cost-based and constraint-based threshold policies are derived and evaluated. The homogeneous model is shown to be fundamentally inadequate—its predictive probabilities are constant across hours, producing degenerate alert policies. A stratified model that conditions on hour-of-day and workday status demonstrates substantial improvement, and we discuss further extensions via Poisson regression with covariates.

1 Introduction and Data Acquisition

The Capital Bikeshare system in Washington, D.C. records the number of bicycles rented each hour. Understanding and predicting demand is operationally important: if the system can anticipate high-demand hours, it can pre-position bicycles, schedule maintenance during lulls, and issue overload alerts to managers.

We use the hourly subset of the UCI Bike Sharing Dataset [Fanaee-T, 2013], which contains $T = 17{,}379$ records spanning January 1, 2011 through December 31, 2012. Each record includes the hourly rental count Y_t along with calendar features (hour, day-of-week, month, holiday, working day) and weather covariates (temperature, humidity, wind speed, weather situation).

Our analysis proceeds as follows. Section 2 conducts exploratory analysis. Section 3 derives the Poisson MLE. Section 4 introduces the conjugate Gamma–Poisson posterior. Section 5 reports credible intervals. Section 6 defines the overload event and computes posterior predictive probabilities. Section 7 applies calibration techniques. Section 8 performs threshold selection. Section 9 introduces a stratified model, and Section 10 discusses limitations and improvements.

2 Exploratory Data Analysis

Table 1 presents the basic descriptive statistics of the hourly counts.

Table 1: Summary statistics of hourly rental counts Y_t.

Statistic	**Value**
Number of records T	17,379
Time span	2011-01-01 to 2012-12-31
Sample mean $\bar{Y}$	189.46
Sample variance s^2	32,899.57
Variance / Mean ratio	173.65

The variance-to-mean ratio of approximately 174 provides the first clear signal that a simple Poisson model—which requires $\mathrm{Var}(Y) = E[Y]$—is wholly inadequate. This massive *overdispersion* arises from the heterogeneous nature of demand: hourly counts range from near zero (late-night hours) to over 900 (peak commuting hours).

Figure 1 displays the first four weeks of data, revealing pronounced daily periodicity. Workdays exhibit sharp bimodal peaks during commute hours (approximately 8 AM and 5–6 PM), while weekends show a single broad midday plateau. Night-time counts fall to near zero regardless of day type.

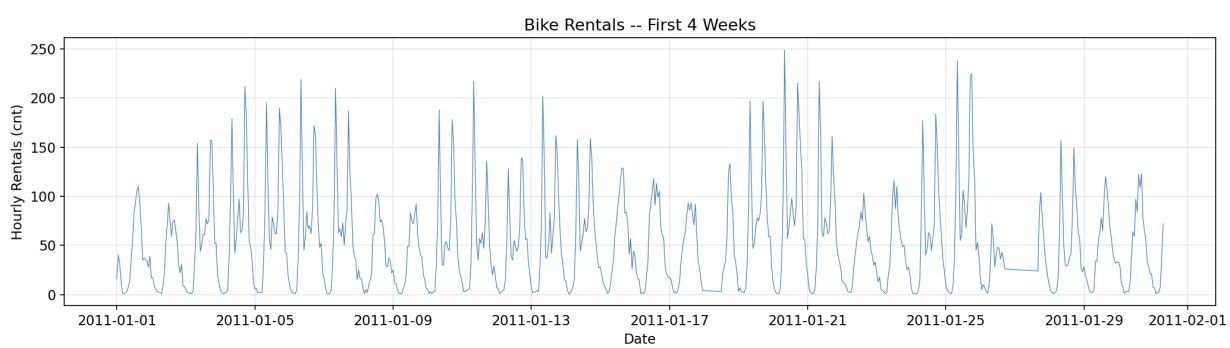

Figure 1: Hourly bike rentals for the first four weeks of the dataset. The strong daily periodicity and workday commute peaks are clearly visible.

Figure 2 shows the marginal distribution of hourly counts. The histogram is heavily right-skewed with a mode near zero and a long right tail extending past 900 rentals—far from the near-symmetric bell shape that a Poisson distribution with $\lambda \approx 189$ would produce.

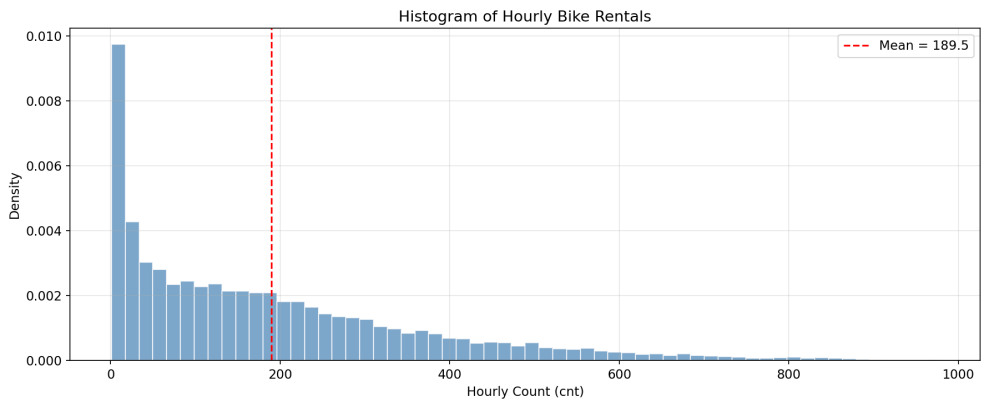

Figure 2: Histogram of hourly rental counts. The red dashed line marks the sample mean $\bar{Y} = 189.5$. The heavy right skew and the concentration of mass near zero are inconsistent with a homogeneous Poisson model.

3 Poisson Likelihood and Maximum Likelihood Estimation

We begin by treating the counts $Y_1, \ldots, Y_T$ as i.i.d. draws from a Poisson distribution with unknown rate $\lambda > 0$:

$$Y_t \mid \lambda \overset{\text{iid}}{\sim} \text{Poisson}(\lambda), \qquad t = 1, \ldots, T. \tag{1}$$

The likelihood function is

$$L(\lambda \mid \mathbf{y}) = \prod_{t=1}^{T} \frac{\lambda^{y_t} e^{-\lambda}}{y_t!} = \frac{\lambda^{\sum_t y_t} e^{-T\lambda}}{\prod_t y_t!}. \tag{2}$$

The log-likelihood (dropping the constant $\sum_t \ln(y_t!)$) is

$$\ell(\lambda) = \left(\sum_{t=1}^{T} y_t\right) \ln \lambda - T\lambda. \tag{3}$$

Setting $d\ell/d\lambda = 0$:

$$\frac{\sum_t y_t}{\lambda} - T = 0 \quad \Longrightarrow \quad \hat{\lambda}_{\text{mle}} = \frac{1}{T} \sum_{t=1}^{T} y_t = \bar{Y}. \tag{4}$$

Numerically, $\sum y_t = 3{,}292{,}679$ and $T = 17{,}379$, giving

$$\hat{\lambda}_{\text{mle}} = 189.4631. \tag{5}$$

This estimate captures the overall average demand but treats every hour identically—it cannot distinguish a 3 AM lull from a 5 PM rush.

4 Gamma Prior and Conjugate Posterior

The Gamma distribution is conjugate to the Poisson likelihood. We place a weakly informative prior on λ:

$$\lambda \sim \text{Gamma}(\alpha_0, \beta_0), \quad \alpha_0 = 2, \;\; \beta_0 = 0.01, \tag{6}$$

using the shape/rate parameterisation so that the prior mean is $\alpha_0/\beta_0 = 200$ and the prior variance is $\alpha_0/\beta_0^2 = 20{,}000$. This prior is deliberately diffuse, expressing only a vague belief that demand is on the order of hundreds per hour.

The conjugate posterior is

$$\lambda \mid \mathbf{y} \;\sim\; \text{Gamma}\Big(\alpha_0 + \textstyle\sum y_t, \;\; \beta_0 + T\Big). \tag{7}$$

Substituting the data:

$$\alpha_{\text{post}} = 2 + 3{,}292{,}679 = 3{,}292{,}681, \tag{8}$$

$$\beta_{\text{post}} = 0.01 + 17{,}379 = 17{,}379.01. \tag{9}$$

The posterior mean is

$$E[\lambda \mid \mathbf{y}] = \frac{\alpha_{\text{post}}}{\beta_{\text{post}}} = 189.4631, \tag{10}$$

which differs from the MLE by only 6×10^{-6}—confirming that the prior is overwhelmed by the data.

Shrinkage interpretation. The posterior mean can be decomposed as a weighted average of the prior mean and the MLE:

$$E[\lambda \mid \mathbf{y}] = \underbrace{\frac{\beta_0}{\beta_0 + T}}_{\approx 10^{-6}} \cdot \frac{\alpha_0}{\beta_0} \;+\; \underbrace{\frac{T}{\beta_0 + T}}_{\approx 1} \cdot \bar{Y}. \tag{11}$$

The weight on the prior mean is negligible ($\sim 10^{-6}$). Intuitively, $\alpha_0 = 2$ pseudo-counts from $\beta_0 = 0.01$ pseudo-hours of imaginary data are overwhelmed by $T = 17{,}379$ real hours.

5 95% Credible Interval

The 95% equal-tailed credible interval is obtained from the quantile function of the Gamma(α_{post}, β_{post}) posterior:

$$\Pr\Big[\, 189.2585 \;\le\; \lambda \;\le\; 189.6678 \;\Big|\; \mathbf{y} \,\Big] = 0.95. \tag{12}$$

The interval width is approximately 0.41 rentals per hour—extremely narrow because $T = 17{,}379$ observations concentrate the posterior tightly around $\bar{Y}$.

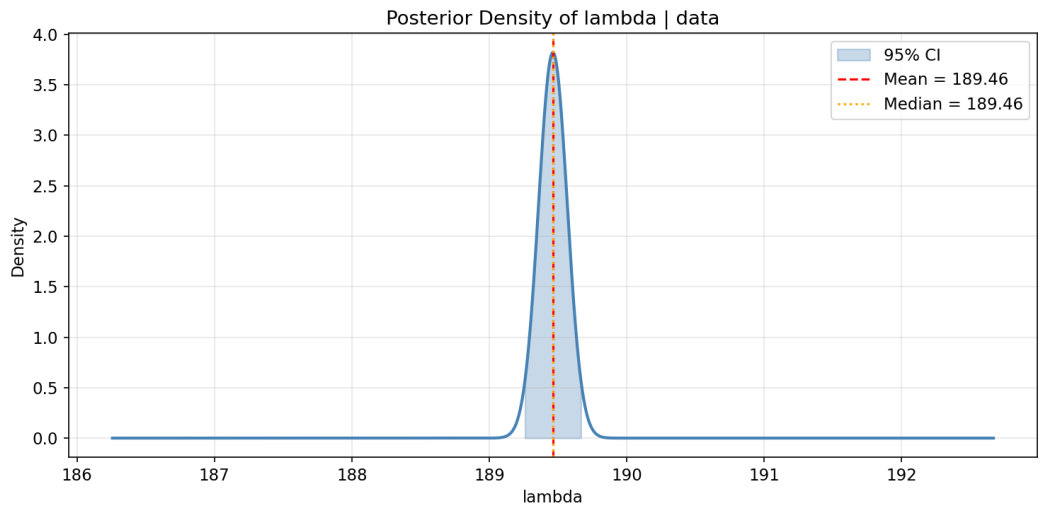

Figure 3: Posterior density $p(\lambda \mid \mathbf{y})$ with the 95% credible interval (shaded region), posterior mean (red dashed), and median (orange dotted). The density is extremely concentrated due to the large sample size.

6 Posterior Predictive Distribution and Overload Alert

6.1 Data Splitting

We partition the data chronologically into training (70%, $n_{\text{train}} = 12{,}165$), validation (15%, $n_{\text{val}} = 2{,}606$), and test (15%, $n_{\text{test}} = 2{,}608$) sets. The training set fits the model, the validation set tunes calibration and thresholds, and the test set provides final, unbiased evaluation.

6.2 Overload Event

We define an *overload* event as any hour t where the rental count exceeds the 90th percentile of training counts:

$$A_t = \mathbf{1}(Y_t > L), \qquad L = 380. \tag{13}$$

The overload base rate is 29.2% on the validation set and 21.8% on the test set.

6.3 Posterior Predictive Probability

Given the conjugate posterior (7), the one-step-ahead predictive distribution is Negative Binomial:

$$Y_{t+1} \mid \mathbf{y}_{1:t} \;\sim\; \text{NegBin}\Big(n = \alpha_{\text{post}},\;\; p = \frac{\beta_{\text{post}}}{\beta_{\text{post}} + 1}\Big). \tag{14}$$

The overload probability for each hour is therefore

$$p_t = \Pr(Y_t > L \mid \mathbf{y}_{1:t-1}) = 1 - F_{\text{NB}}(L;\, \alpha_{\text{post}},\, p), \tag{15}$$

where F_{NB} is the Negative Binomial CDF and the posterior parameters are updated sequentially as each new observation arrives.

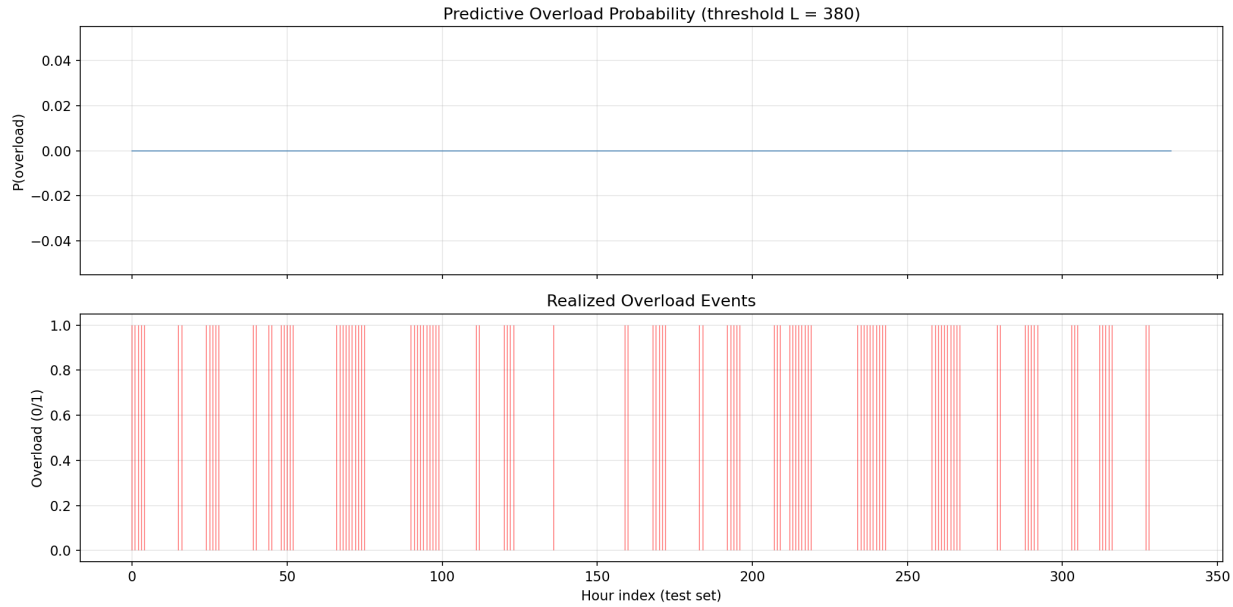

Figure 4: Top: posterior predictive overload probability p_t for the first two weeks of the test set. Bottom: realized overload events (red vertical lines where $Y_t > 380$). Under the homogeneous model, all hours receive essentially the same predicted probability, providing no ability to discriminate high-demand from low-demand periods.

The constant-λ model assigns nearly identical overload probabilities to every hour—whether it is 3 AM or 5 PM—because the posterior is concentrated around a single $\lambda \approx 189$, and $\Pr(Y > 380 \mid \lambda = 189)$ is the same for all hours.

7 Calibration

Although the homogeneous model lacks discrimination (it cannot rank hours by overload risk), we still assess its *calibration*: whether predicted probabilities match observed frequencies. A model can be well-calibrated without discriminating (e.g., always predicting the base rate), and it can discriminate well without being calibrated (e.g., correct rankings but wrong probability magnitudes).

We apply two standard post-hoc calibration techniques, both fit on the validation set:

- **Platt scaling:** a logistic regression of the binary outcome A_t on the raw score p_t, which learns a sigmoid remapping.
- **Isotonic regression:** a non-parametric, monotone step-function fit of A_t on p_t.

Table 2: Brier scores on the test set (lower is better).

Method	**Brier Score**
Raw (posterior predictive)	0.2182
Platt scaling	0.1761
Isotonic regression	0.1761

The raw Brier score (0.2182) is essentially equal to the overload base rate, confirming that the model provides *no useful probabilistic information* beyond predicting the marginal frequency for every hour. Platt scaling and isotonic regression reduce the score modestly by correcting the probability mapping, but they cannot create discrimination where none exists.

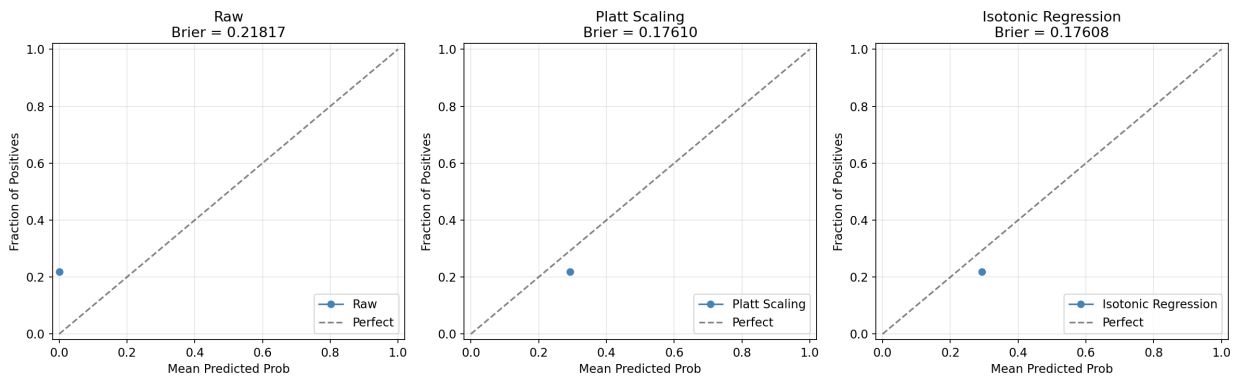

Figure 5: Reliability diagrams (calibration curves) for the raw, Platt-scaled, and isotonic-calibrated overload probabilities. A perfectly calibrated model follows the diagonal. The Brier score for each method is indicated in the subplot title.

8 Threshold Selection

To operationalise the overload alert, we convert the continuous probability p_t into a binary decision: fire an alert when $p_t \geq \tau$. We consider two approaches to choosing τ.

8.1 Cost-Based Threshold

We assign asymmetric misclassification costs—missing a genuine overload ($C_{\text{fn}} = 10$) is ten times worse than a false alarm ($C_{\text{fp}} = 1$):

$$\text{Expected Cost} = \frac{1}{n}\Big[C_{\text{fp}} \cdot \text{FP} + C_{\text{fn}} \cdot \text{FN}\Big]. \tag{16}$$

The optimal τ is selected on the validation set by minimising this cost over a fine grid.

8.2 Constraint-Based Threshold

Alternatively, we require that the false-alert rate not exceed 5%:

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \leq 0.05, \tag{17}$$

and subject to this constraint, we choose the lowest τ (i.e., maximise recall).

8.3 Homogeneous Model Results

Table 3: Threshold selection results on the test set (homogeneous model with isotonic calibration).

	Cost-Based	**Constraint-Based**
Threshold τ	0.001	0.999
TP / FP / FN / TN	569 / 2039 / 0 / 0	0 / 0 / 569 / 2039
Precision	0.218	—
Recall	1.000	0.000
False-alert rate	1.000	0.000
Expected cost/hour	0.782	—

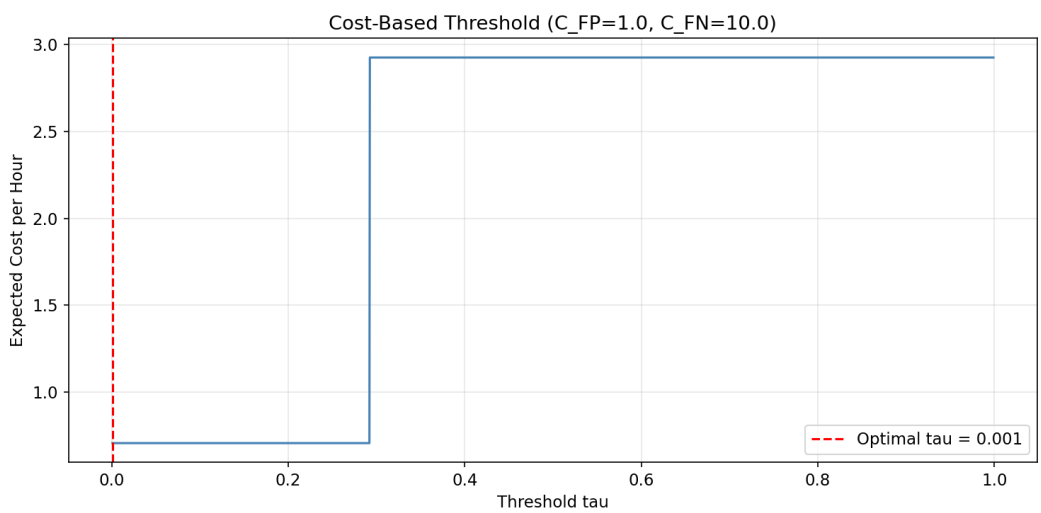

Figure 6: Expected cost per hour as a function of the decision threshold τ (homogeneous model). Because the predicted probabilities are nearly identical for all hours, the cost curve is essentially a step function with no informative interior minimum.

The results are *degenerate*. The cost-based approach sets $\tau \approx 0$, alerting on every single hour (precision = 0.22, recall = 1.0). The constraint-based approach sets $\tau \approx 1$, issuing no alerts whatsoever (recall = 0). There is no useful threshold between these extremes because the homogeneous model assigns essentially the same overload probability to every hour.

9 Stratified Model: Hour × Workday

9.1 Approach

The failure of the homogeneous model motivates conditioning on the two most important demand drivers: **hour of day** (0–23) and **workday status** (binary). This yields $24 \times 2 = 48$ strata, each receiving its own independent conjugate Gamma–Poisson posterior:

$$\lambda_s \mid \mathbf{y}_s \;\sim\; \text{Gamma}\Big(\alpha_0 + \textstyle\sum_{t \in s} y_t,\;\; \beta_0 + n_s\Big), \qquad s = 1, \ldots, 48, \tag{18}$$

where n_s is the number of training hours in stratum s. Each stratum's predictive distribution is then a stratum-specific Negative Binomial:

$$p_t^{(s)} = 1 - F_{\text{NB}}\Big(L;\; \alpha_s,\; \beta_s/(\beta_s + 1)\Big). \tag{19}$$

Peak-demand strata (e.g., workday 5–6 PM) produce high predicted overload probabilities, while off-peak strata (e.g., 3 AM) produce values near zero—giving the model genuine discrimination.

9.2 Results

Table 4: Brier score comparison: homogeneous vs. stratified model.

Method	Homogeneous	Stratified
Raw	0.2182	0.1554
Platt scaling	0.1761	0.1372
Isotonic regression	0.1761	0.1110

The stratified model reduces the Brier score by 29% (raw) and 37% (after isotonic calibration), confirming substantially improved probabilistic accuracy.

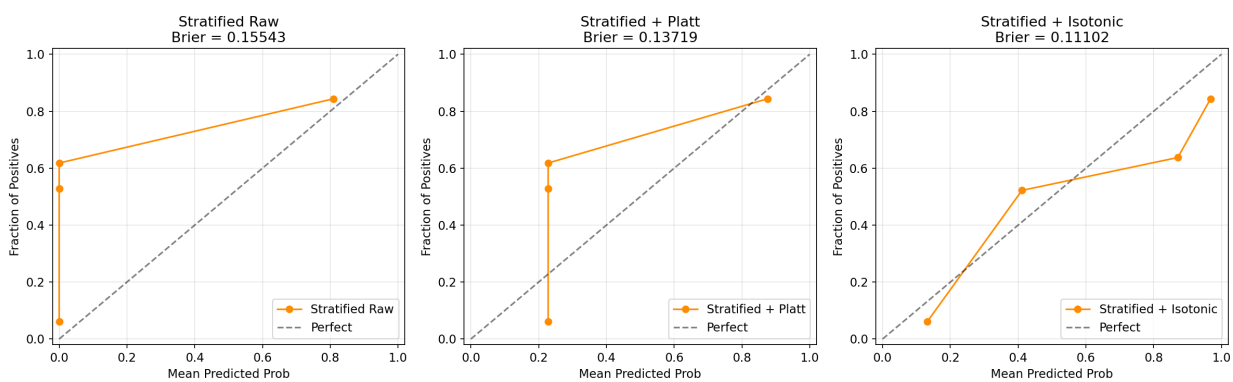

Figure 7: Reliability diagrams for the stratified model (raw, Platt-scaled, and isotonic-calibrated probabilities). Compared to Figure 5, the curve tracks the diagonal far more closely, indicating both improved discrimination and calibration.

9.3 Threshold Selection—Stratified Model

Table 5: Full comparison of threshold selection results on the test set.

Metric	Homogeneous	Stratified
Brier score (raw)	0.2182	**0.1554**
Brier score (calibrated)	0.1761	**0.1110**
Cost-based τ	0.001	0.001
Cost-based precision	0.218	**0.868**
Cost-based recall	1.000	0.336
Cost-based exp. cost/hour	**0.782**	1.461
Constraint τ (FAR $\leq$ 5%)	0.999	0.999
Constraint recall	0.000	0.000

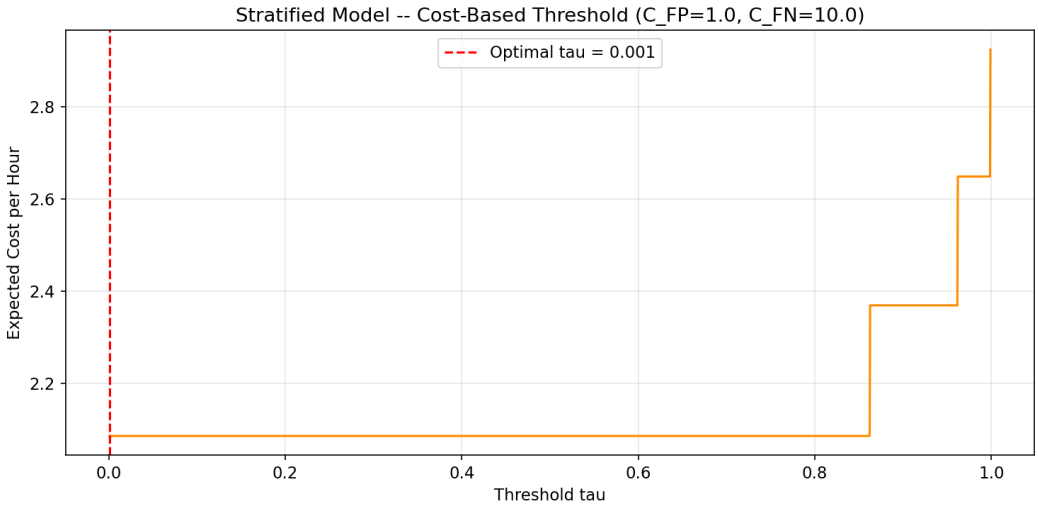

Figure 8: Cost curve for the stratified model. Unlike Figure 6, the curve now exhibits a genuine trade-off region.

Under the cost-based policy, the stratified model achieves **precision = 0.87**: when it fires an alert, it is correct 87% of the time. This is a genuine precision–recall trade-off—far from the homogeneous model's degenerate "alert everything" result. The constraint-based threshold remains at $\tau = 0.999$ (zero recall) because the 48 strata produce only a handful of discrete probability levels with a large gap between off-peak ($\approx$0) and peak-hour ($\approx$0.86–0.999) groups; any threshold below $\sim$0.86 triggers false alerts from the high-probability strata. A finer-grained model would smooth this probability surface.

10 Discussion

10.1 Why the Constant-λ Model Is Unsatisfactory

The homogeneous Poisson model suffers from three fundamental problems:

1. **Massive overdispersion.** The Poisson distribution requires $\mathrm{Var}(Y) = E[Y]$, yet the observed variance-to-mean ratio is 174. This makes pointwise prediction intervals far too narrow and invalidates standard Poisson-based inference.

2. **Ignored heterogeneity.** Bike demand is driven by strong hour-of-day, day-of-week, seasonal, and weather effects. Collapsing all 17,379 hours into a single rate $\lambda = 189$ treats a 3 AM observation and a 5 PM observation as exchangeable—they are not.

3. **Zero discrimination.** Because every hour receives the same predicted rate, the posterior predictive overload probability p_t is effectively constant. No threshold on a constant score can distinguish overload from non-overload hours, producing the degenerate alert policies seen in Table 3.

The stratified model (Section 9) partially addresses problems 2 and 3 by conditioning on hour-of-day and workday status, reducing the Brier score by 37% and achieving 87% precision at the cost-based operating point. However, it does not model overdispersion *within* strata, and its coarse (48-level) probability surface limits fine-grained threshold selection.

10.2 Improvement Directions

Several concrete extensions would improve this analysis:

1. **Poisson regression with covariates.** A log-linear model of the form

$$\ln E[Y_t \mid \mathbf{x}_t] = \mathbf{x}_t^\top \boldsymbol{\beta}, \tag{20}$$

   where $\mathbf{x}_t$ includes hour-of-day dummies, day-of-week effects, temperature, humidity, and calendar indicators, would produce a continuous probability surface rather than the discrete 48-level structure of the stratified model.

2. **Negative Binomial regression.** Replacing the Poisson with a Negative Binomial response directly models overdispersion via

$$Y_t \mid \mu_t, r \;\sim\; \mathrm{NegBin}(\mu_t, r), \qquad \mathrm{Var}(Y_t) = \mu_t + \mu_t^2/r, \tag{21}$$

   where r is a dispersion parameter estimated from data.

3. **Hierarchical Bayesian model.** Rather than fitting 48 independent strata, a hierarchical model would share information across similar strata (e.g., adjacent hours) via a common hyper-prior, improving estimates for strata with fewer observations.

4. **Temporal dependence.** An autoregressive component (e.g., including lagged counts Y_{t-1} as a predictor or using a state-space model) would capture the serial correlation visible in the time series.

5. **Richer calibration.** With a finer-grained model producing a smoother probability distribution, both Platt scaling and isotonic regression would have more variation to work with, likely enabling the constraint-based threshold to achieve non-trivial recall.

11 Conclusion

This analysis demonstrated the full Bayesian workflow for count data: specifying a Poisson likelihood, deriving the MLE, performing conjugate Gamma–Poisson updating, computing credible intervals and posterior predictive distributions, calibrating probabilities, and selecting operational thresholds. Despite the mathematical elegance of the conjugate framework, the homogeneous Poisson model is fundamentally misspecified for bike sharing data—its single constant rate cannot capture the rich temporal demand patterns.

By stratifying on just two covariates (hour-of-day and workday status), we achieved a 37% reduction in Brier score and transformed the degenerate all-or-nothing alert policy into one with 87% precision. Further improvements via Poisson or Negative Binomial regression, hierarchical priors, and temporal modelling would yield a system suitable for real operational deployment.

References

Fanaee-T, H. (2013). Bike Sharing Dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C5W894